

# Improving Methodological Research in Psychology: Preregistration, Transparent Reporting, and Benchmarking

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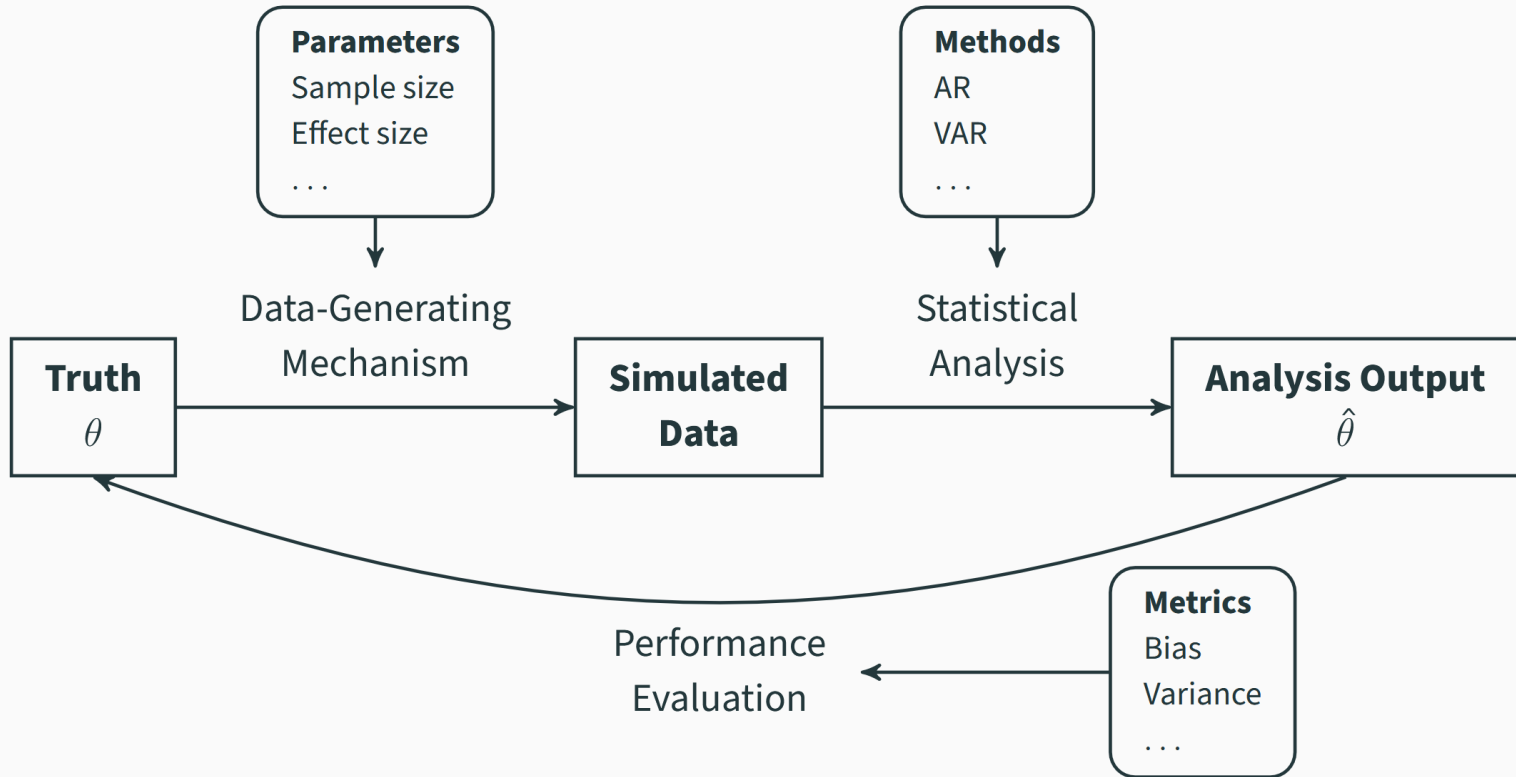
Leibniz-Institut für Bildungsverläufe · June 2026

Björn Siepe

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# Part I: Background and Issues

# Simulation studies



# Issues in simulation studies



Marcin Wichary - Wikimedia

“ *Statisticians ... often pay too little attention to their own principles of design*  
Hoaglin & Andrews, 1975

# Meta-Science on Simulation Studies

## Interdisciplinary

- (Bio-)statistics: Morris et al. (2019)
- Ecology: Williams et al. (2024)
- Psychometrics: Feinberg & Rubright (2016)

## Transparency

- Issues similar to other empirical research (Boulesteix et al., 2020)
- Insufficient reporting standards (e.g., Hoaglin and Andrews, 1975)
- Reproducibility? (e.g., Luijken et al., 2023)

## Execution Issues

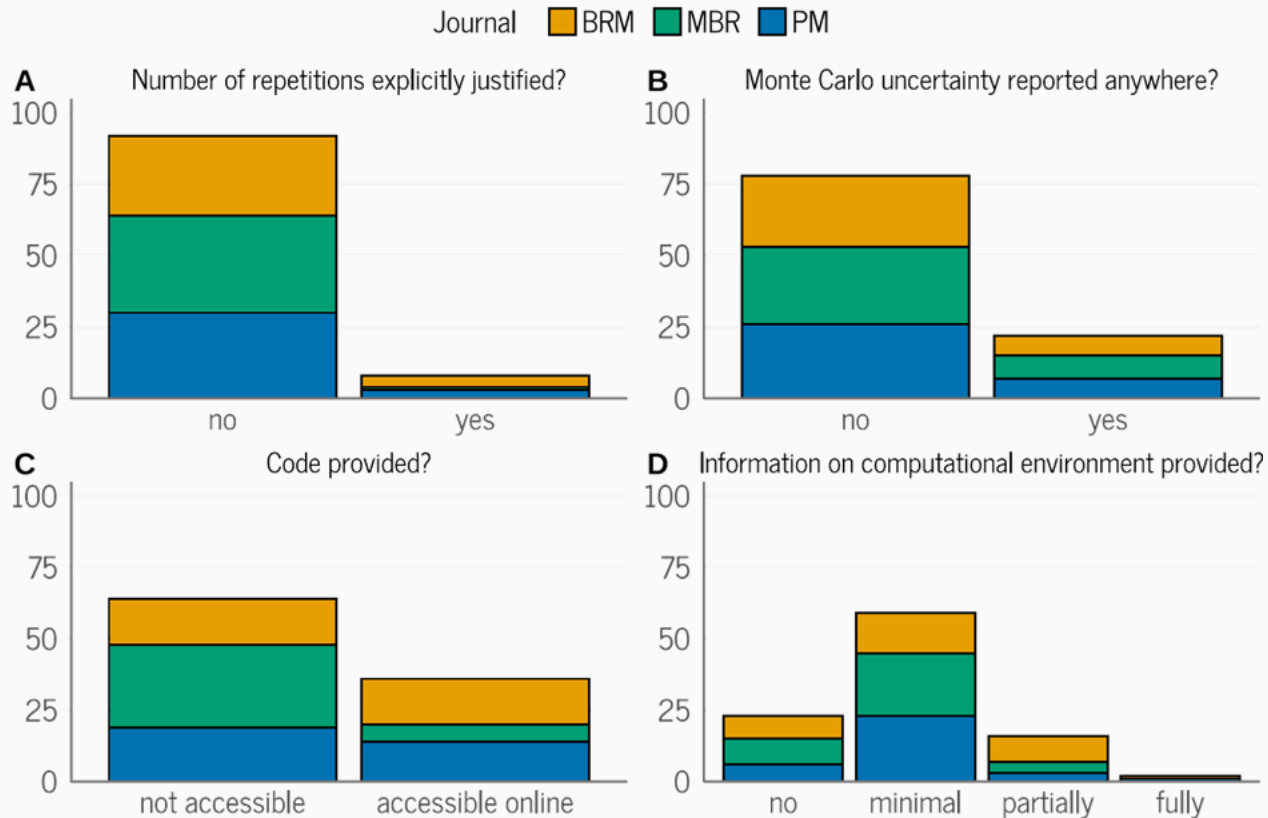
- Many degrees of freedom (Pawel et al., 2024)
- Over-Optimism (e.g., Ullmann et al., 2022)
- Unclear handling of non-convergence (Pawel et al., 2025; Wünsch et al., 2025)



## Simulation Studies for Methodological Research in Psychology: A Standardized Template for Planning, Preregistration, and Reporting

Björn S. Siepe<sup>1</sup>, František Bartoš<sup>2</sup>, Tim P. Morris<sup>3</sup>, Anne-Laure Boulesteix<sup>4, 5</sup>,  
Daniel W. Heck<sup>1</sup>, and Samuel Pawel<sup>6, 7</sup>

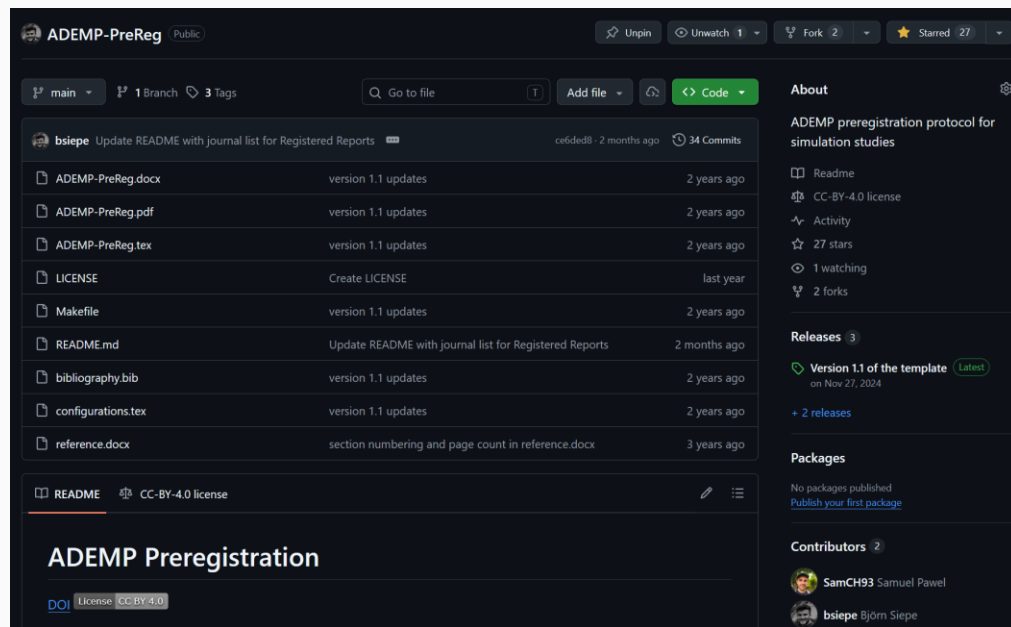
# Simulation studies in psychology



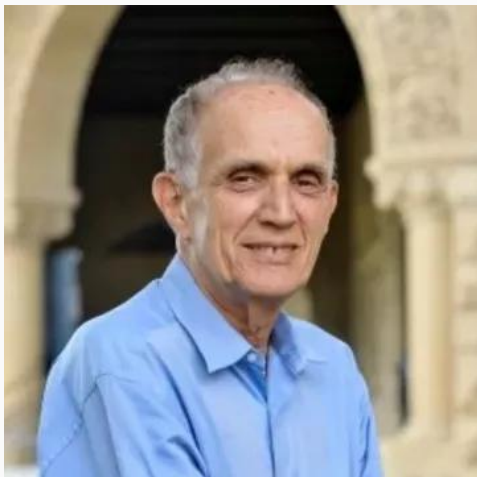
# Simulation studies in psychology

| Performance measure           | Definition                                               | Estimate                                                                                                 | MCSE                                                                            | $n_{\text{sim}}$                                                            |
|-------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Bias                          | $E(\hat{\theta}) - \theta$                               | $(\sum_{i=1}^{n_{\text{sim}}} \hat{\theta}_i / n_{\text{sim}}) - \theta$                                 | $\sqrt{S_{\hat{\theta}}^2 / n_{\text{sim}}}$                                    | $S_{\hat{\theta}}^2 / \text{MCSE}_{*}^2$                                    |
| Relative bias                 | $\{E(\hat{\theta}) - \theta\} / \theta$                  | $\{(\sum_{i=1}^{n_{\text{sim}}} \hat{\theta}_i / n_{\text{sim}}) - \theta\} / \theta$                    | $\sqrt{S_{\hat{\theta}}^2 / (\theta^2 n_{\text{sim}})}$                         | $S_{\hat{\theta}}^2 / (\text{MCSE}_{*}^2 \theta^2)$                         |
| MSE                           | $E\{(\hat{\theta} - \theta)^2\}$                         | $\sum_{i=1}^{n_{\text{sim}}} (\hat{\theta}_i - \theta)^2 / n_{\text{sim}}$                               | $\sqrt{S_{(\hat{\theta}-\theta)^2}^2 / n_{\text{sim}}}$                         | $S_{(\hat{\theta}-\theta)^2}^2 / \text{MCSE}_{*}^2$                         |
| RMSE                          | $\sqrt{E\{(\hat{\theta} - \theta)^2\}}$                  | $\sqrt{\sum_{i=1}^{n_{\text{sim}}} (\hat{\theta}_i - \theta)^2 / n_{\text{sim}}}$                        | $\sqrt{S_{(\hat{\theta}-\theta)^2}^2 / (4n_{\text{sim}} \widehat{\text{MSE}})}$ | $S_{(\hat{\theta}-\theta)^2}^2 / (4\widehat{\text{MSE}} \text{MCSE}_{*}^2)$ |
| Empirical variance            | $\text{Var}(\hat{\theta})$                               | $S_{\hat{\theta}}^2$                                                                                     | $S_{\hat{\theta}}^2 \sqrt{2 / (n_{\text{sim}} - 1)}$                            | $1 + 2(S_{\hat{\theta}}^2)^2 / \text{MCSE}_{*}^2$                           |
| Empirical standard error      | $\sqrt{\text{Var}(\hat{\theta})}$                        | $\sqrt{S_{\hat{\theta}}^2}$                                                                              | $\sqrt{S_{\hat{\theta}}^2 / \{2(n_{\text{sim}} - 1)\}}$                         | $1 + S_{\hat{\theta}}^2 / (2\text{MCSE}_{*}^2)$                             |
| Coverage                      | $\Pr(\text{CI includes } \theta)$                        | $\sum_{i=1}^{n_{\text{sim}}} \mathbb{1}(\text{CI}_i \text{ includes } \theta) / n_{\text{sim}}$          | $\sqrt{\widehat{\text{Cov}}(1 - \widehat{\text{Cov}}) / n_{\text{sim}}}$        | $\widehat{\text{Cov}}(1 - \widehat{\text{Cov}}) / \text{MCSE}_{*}^2$        |
| Power (or Type I error rate)  | $\Pr(\text{Test rejects } H_0)$                          | $\sum_{i=1}^{n_{\text{sim}}} \mathbb{1}(\text{Test}_i \text{ rejects } H_0) / n_{\text{sim}}$            | $\sqrt{\widehat{\text{Pow}}(1 - \widehat{\text{Pow}}) / n_{\text{sim}}}$        | $\widehat{\text{Pow}}(1 - \widehat{\text{Pow}}) / \text{MCSE}_{*}^2$        |
| Mean CI width                 | $E(\text{CI}_{\text{upper}} - \text{CI}_{\text{lower}})$ | $\sum_{i=1}^{n_{\text{sim}}} (\text{CI}_{i,\text{upper}} - \text{CI}_{i,\text{lower}}) / n_{\text{sim}}$ | $\sqrt{S_W^2 / n_{\text{sim}}}$                                                 | $S_W^2 / \text{MCSE}_{*}^2$                                                 |
| Mean of generic statistic $G$ | $E(G)$                                                   | $\sum_{i=1}^{n_{\text{sim}}} G_i / n_{\text{sim}}$                                                       | $\sqrt{S_G^2 / n_{\text{sim}}}$                                                 | $S_G^2 / \text{MCSE}_{*}^2$                                                 |

1. Instructions
2. General information
3. **Aims**
4. **Data-generating mechanisms**
5. **Estimands & targets**
6. **Methods**
7. **Performance measures**
8. **Computational details**



# Issues in simulation studies



<https://statistics.stanford.edu/people/bradley-efron>

“[...] it is *very difficult to run an honest simulation comparison, and easy to inadvertently cheat by choosing favorable examples, or by not putting as much effort into optimizing the dull old standard as the exciting new challenger.*

Brad Efron, 2001

# Potential Solutions

## Adversarial Collaboration



The Splintered Mind Blog

## Analysis Blinding



Artwork by Sandbox Studio, Chicago with Corinne Mucha

## Reporting Standards/Full Disclosure/Lab Handbook

**ADEMP-PreReg  
Template for Simulation Studies**

# **Part II:** To preregister or not to preregister?

# The idea of simulation protocols



Research Article

## The design of simulation studies in medical statistics

[Andrea Burton](#) ✉, [Douglas G. Altman](#), [Patrick Royston](#), [Roger L. Holder](#)

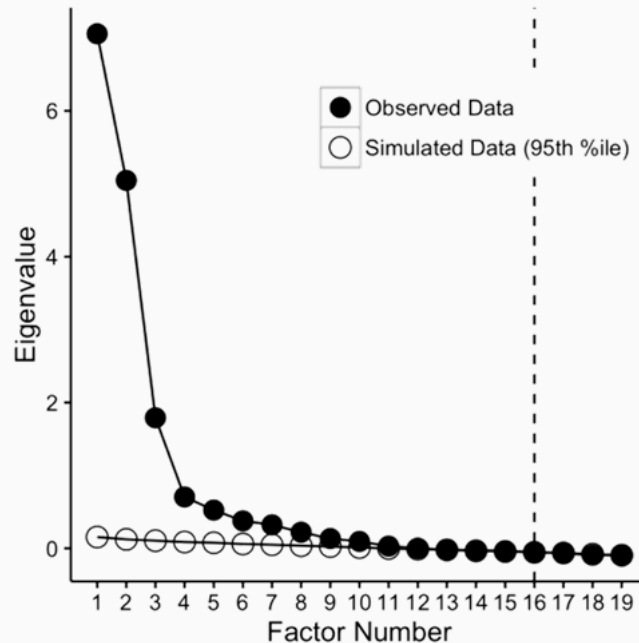
First published: 31 August 2006 | <https://doi.org/10.1002/sim.2673> | [VIEW METRICS](#)

*“When planning a simulation study, it is recommended that a detailed protocol be produced, giving full details of how the study will be performed, analysed and reported.”*

Burton et al., 2006

# Claims based on simulation studies

Add  
citation  
of paper

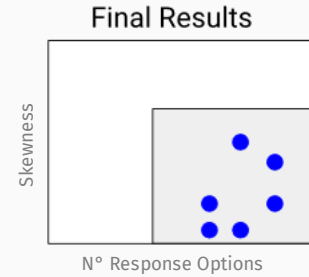


Sakaluk & Short (2016)

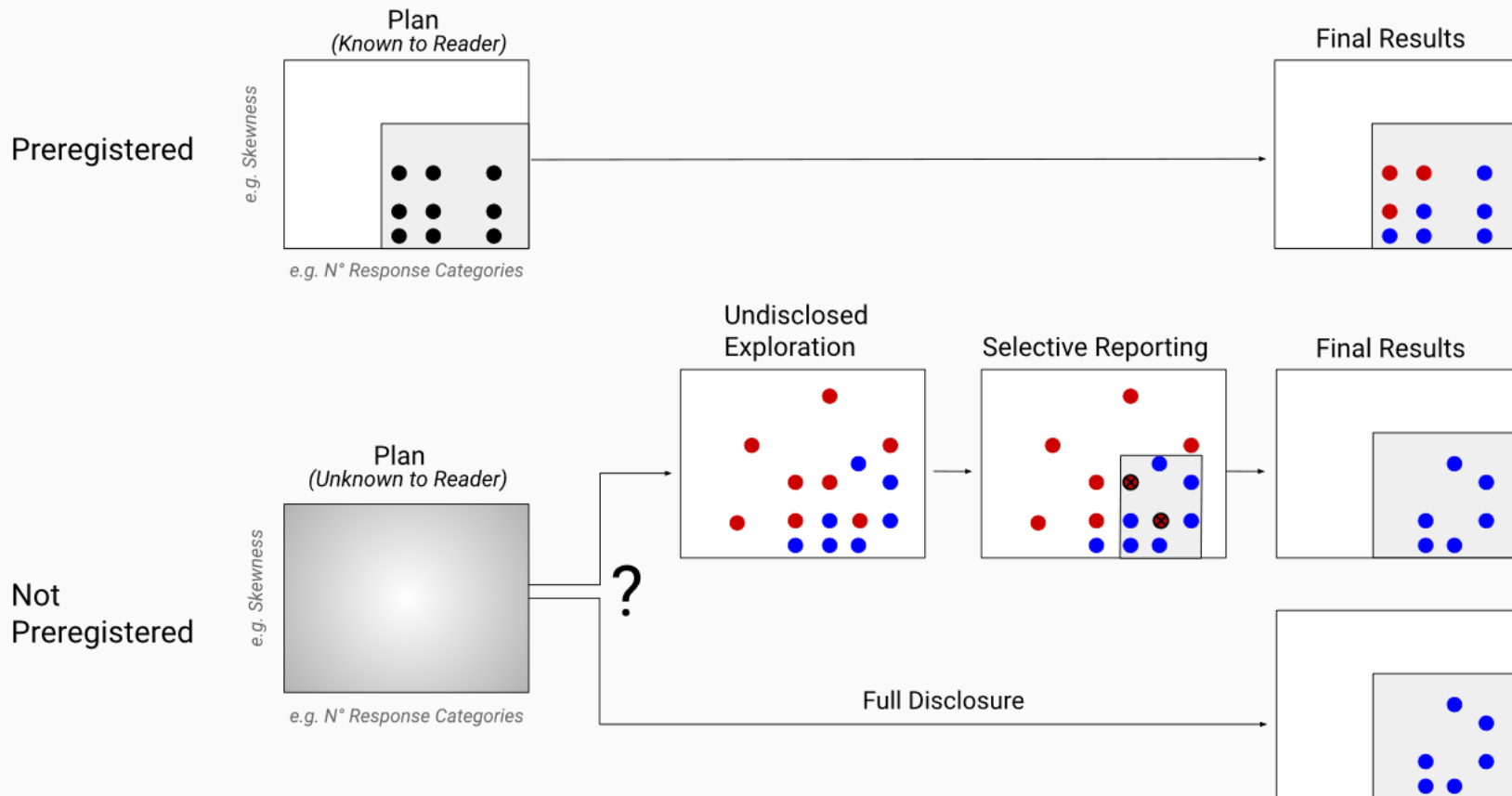
“*We recommend parallel analysis for ordinal items*

Hypothetical authors

# Illustration of preregistration



# Illustration of preregistration



# Theoretical advantages of preregistration



Preregistration reduces **epistemic uncertainty** about the process that produced simulation results

## **Falsificationist**

Simulation only counts as evidence if it could have caught the method failing

## **Inductionist**

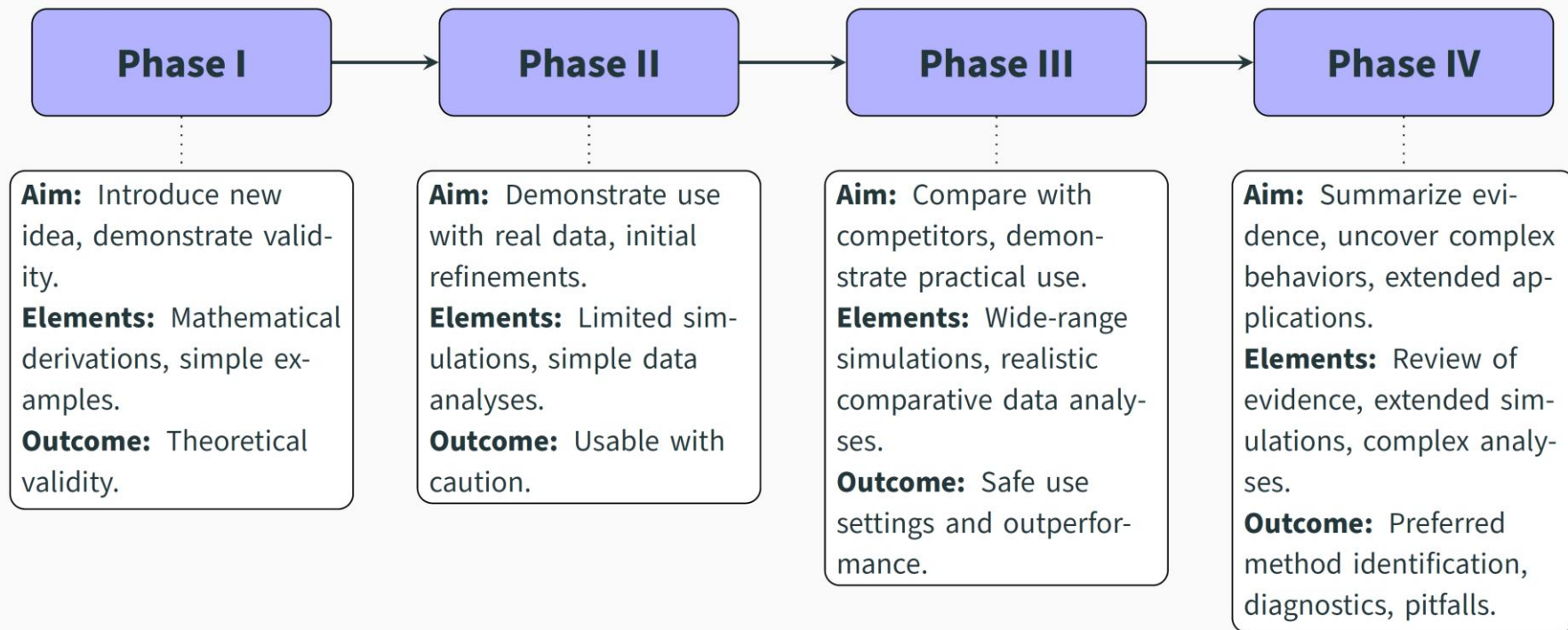
Convincing simulations need difficult conditions (catch a bad method) and easy enough conditions (let a good method succeed)

## **Predictivist**

Predicting an outcome and prespecifying a plan in advance carries more evidential weight than explaining it afterward

# When might they work?

Increasing usefulness of preregistration →



# FAQ about preregistration



**Is preregistration in itself a sign of good quality?**

**What about deviations?**

**Do I need theoretical expectations?**

**When does a simulation study start?**

**What about cheating?**

No, but allows assessment thereof

Be transparent

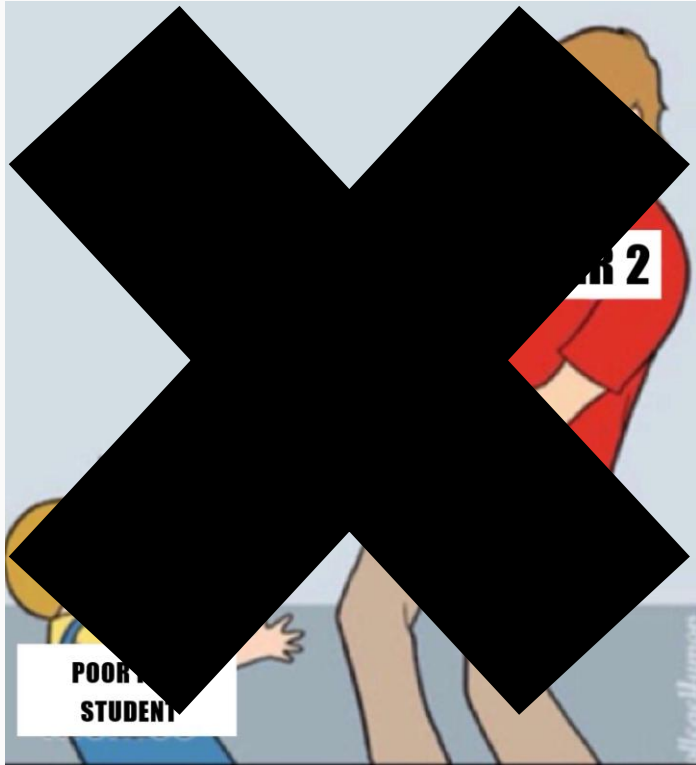
No, but can be helpful

Hard to determine, but there are practical tools

Can never be fully ruled out



# Registered Reports to the rescue



Created with ImageResizer

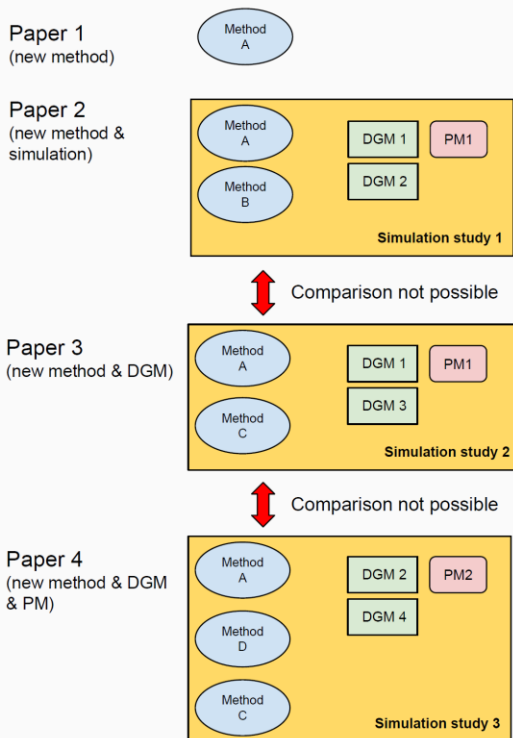


Centre for Open Science

# Part III: Benchmarking

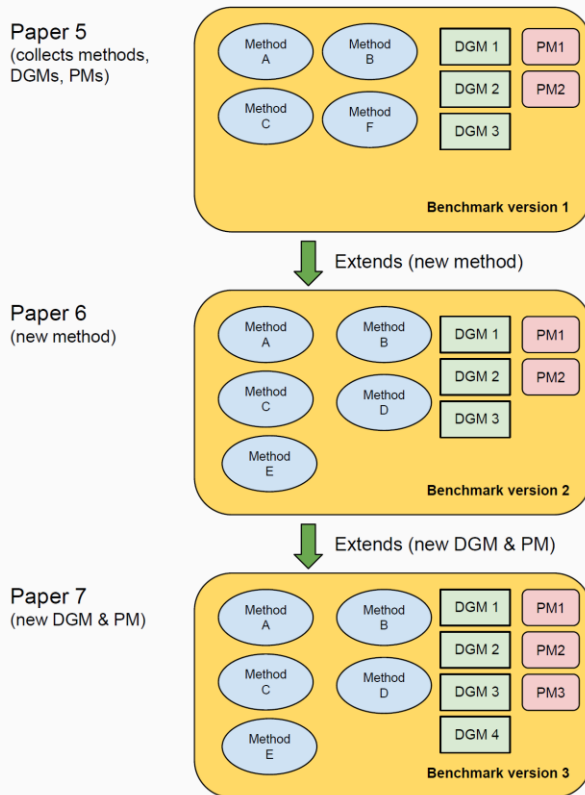
# Incentivizing cumulative simulation research

## Separate Studies (Status Quo)



DGM: Data-Generating Mechanism  
PM: Performance Measure

## Continuous Synthetic Benchmarking (Proposal)



## **Living Synthetic Benchmarks: A Neutral and Cumulative Framework for Simulation Studies**

František Bartoš <sup>1</sup>, Samuel Pawel <sup>2</sup>, Björn S. Siepe <sup>3</sup>

# Synthetic benchmarking for simulation studies

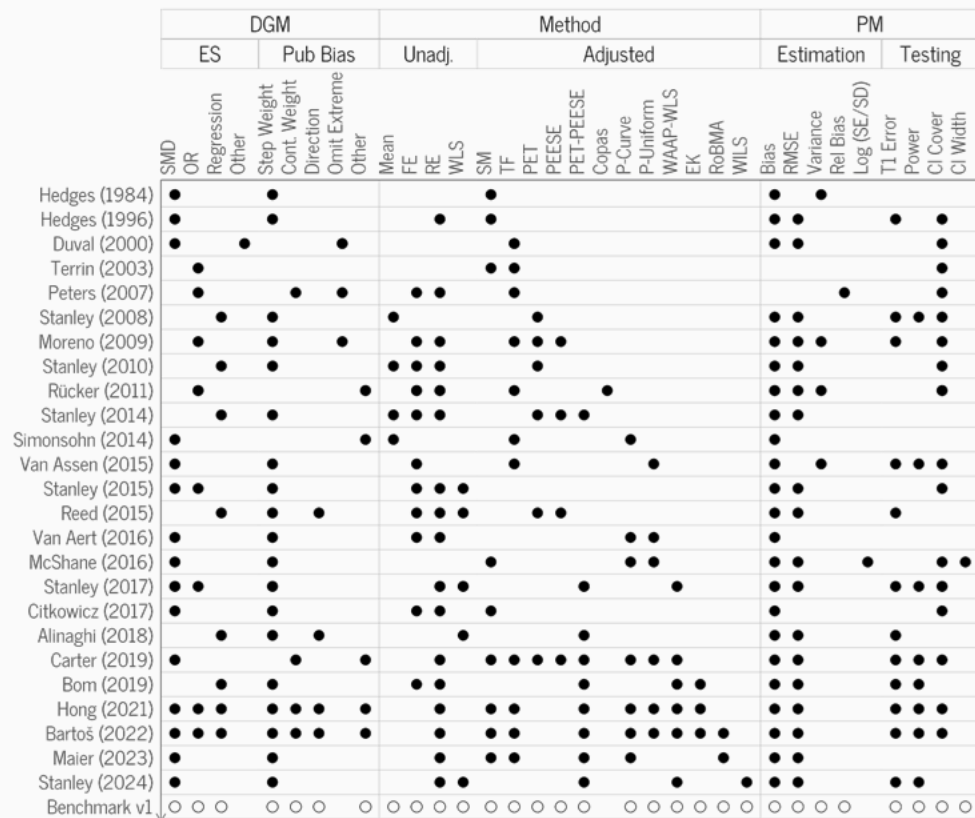


Figure 3: Timeline of major simulation studies on publication bias adjustment

# Synthetic benchmarking for simulation studies

[fbartos.github.io/PublicationBiasBenchmark/](https://fbartos.github.io/PublicationBiasBenchmark/)

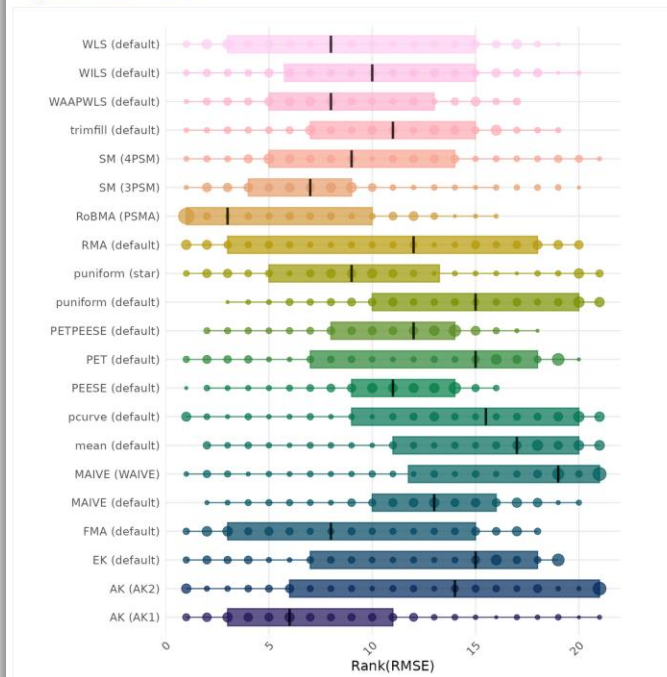
## By-Condition Performance (Conditional on Method Convergence)

The results below are conditional on method convergence. Note that the methods might differ in convergence rate and are therefore not compared on the same data sets.

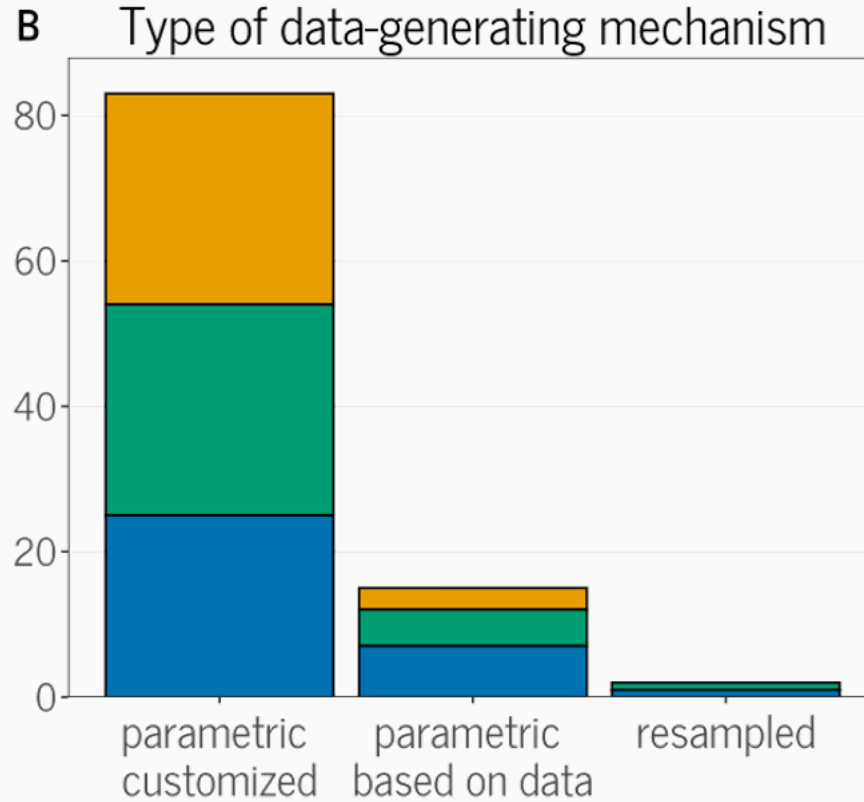
Convergence **RMSE** Bias Empirical SE Interval Score 95% CI Coverage

95% CI Width Log Positive Likelihood Ratio Log Negative Likelihood Ratio

Type I Error Rate Power

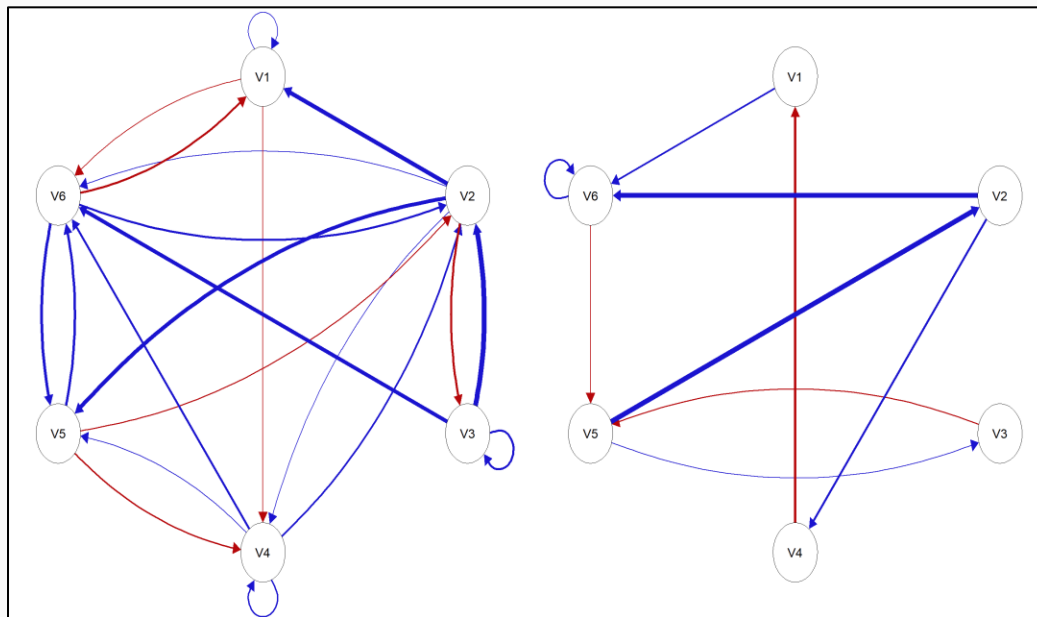


# What about *real* benchmarking?



# Particular Issues in Psychology

What is a realistic data-generating mechanism?



Hard to understand when performance differences actually matter

# Real-data benchmarking

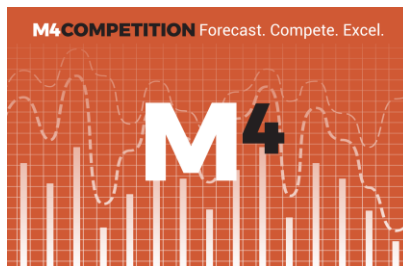
**Idea:** Application of different algorithms to real-world data

## Computer Vision/ML



Source: Steni Sebastian, Medium

## Forecasting



Source: Rob Hyndman

## Biostatistics

Robinson and Vitek *Genome Biology* (2019) 20:205  
<https://doi.org/10.1186/s13059-019-1846-5>

### EDITORIAL

## Benchmarking comes of age

Mark D. Robinson<sup>1\*</sup> and Olga Vitek<sup>2\*</sup>

Psychology has been slow to adopt benchmarking (Rocca & Yarkoni, 2021)

# Introducing openESM



## Database

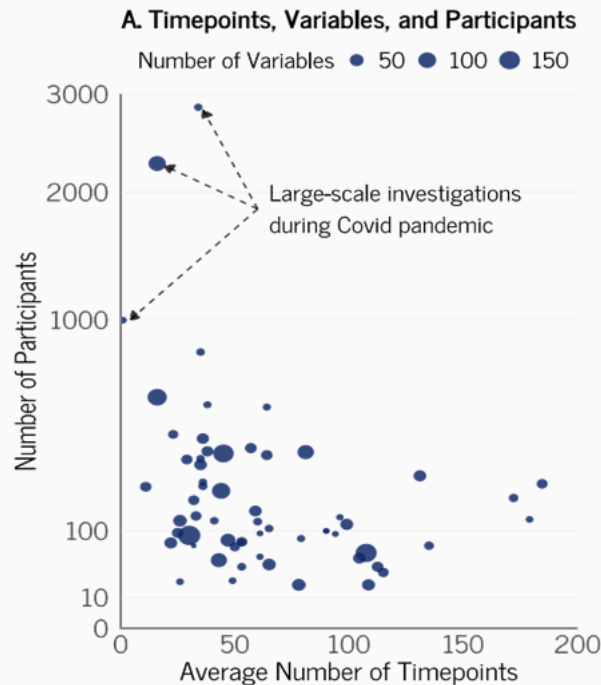
Open-source  
Harmonized  
Rich metadata

## Scale

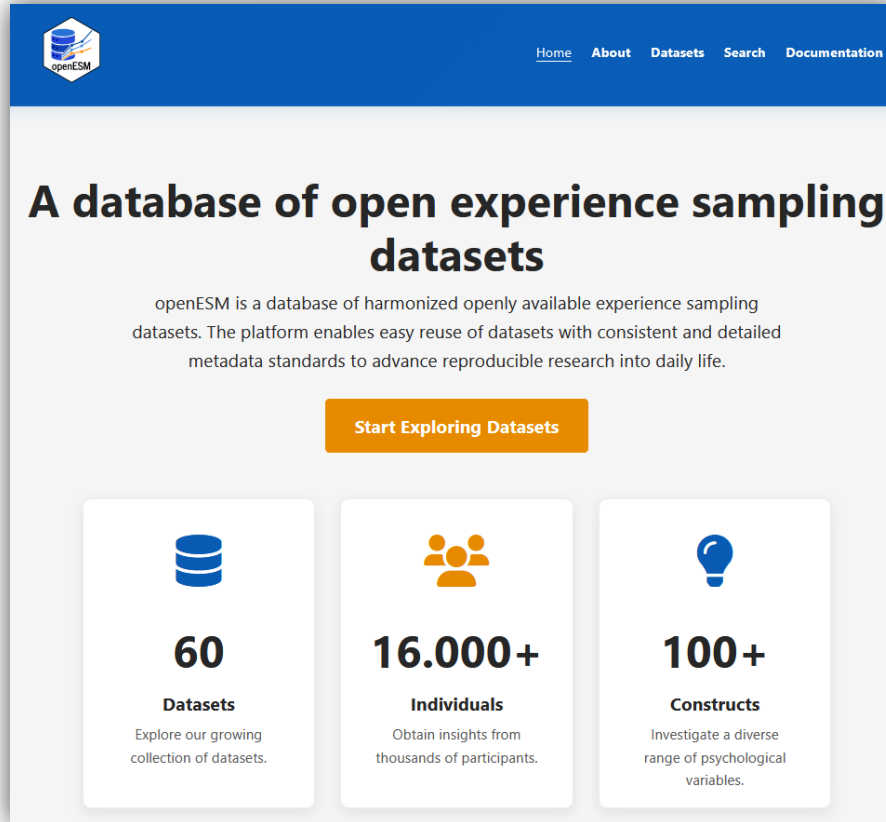
**60** datasets  
**>16,000** participants  
**>740,000** observations

## Access

Website with search  
 and  packages



Siepe, B. S., Haslbeck, J. M. B., Kloft, M., Büchner, A., Zhang, Y., Fried, E. I., & Heck, D. W. (2025)



The screenshot shows the homepage of the openESM website. At the top is a blue navigation bar with the openESM logo on the left and links for Home, About, Datasets, Search, and Documentation on the right. The main content area has a light gray background. It features a large heading 'A database of open experience sampling datasets' followed by a paragraph describing the database. Below this is an orange button labeled 'Start Exploring Datasets'. At the bottom, there are three white cards, each with an icon, a large number, a title, and a short description.

**openESM**

[Home](#) [About](#) [Datasets](#) [Search](#) [Documentation](#)

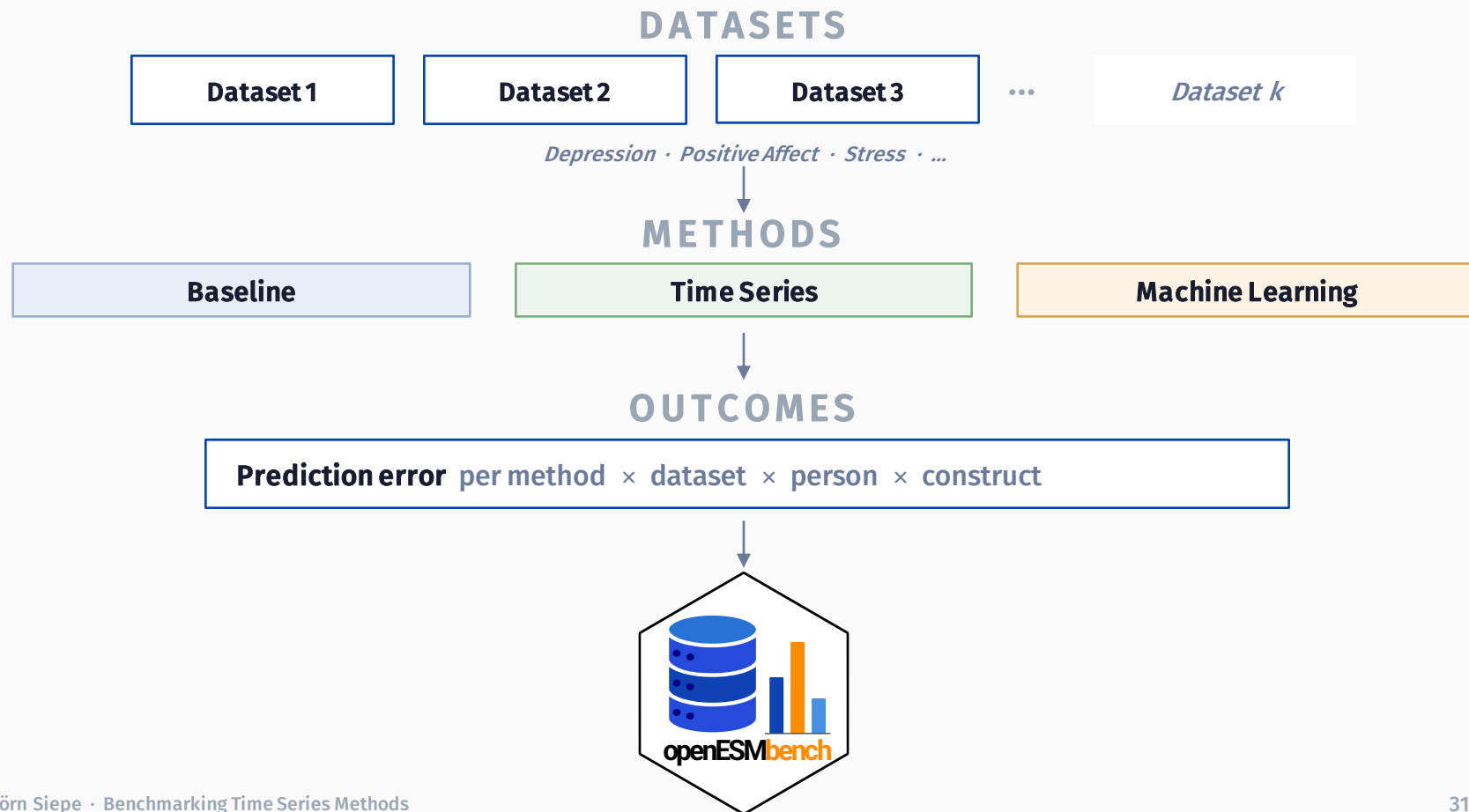
## A database of open experience sampling datasets

openESM is a database of harmonized openly available experience sampling datasets. The platform enables easy reuse of datasets with consistent and detailed metadata standards to advance reproducible research into daily life.

[Start Exploring Datasets](#)

| Icon                                                                                | Count   | Category    | Description                                             |
|-------------------------------------------------------------------------------------|---------|-------------|---------------------------------------------------------|
|    | 60      | Datasets    | Explore our growing collection of datasets.             |
|   | 16.000+ | Individuals | Obtain insights from thousands of participants.         |
|  | 100+    | Constructs  | Investigate a diverse range of psychological variables. |

# Benchmarking psychological time series methods



Methodological research has **quality issues** like any empirical field

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Two potential solutions:

**Preregistration**  
makes generalizability  
assessable and process  
transparent, **not**  
**automatically better**

**Benchmarking**  
closes the gap in **two ways**:

- explicit conditions
- real-world grounding

Item Response Warehouse



# Collaborators



# Thank You!

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<https://openesmdata.org>



[bjoernsiepe@gmail.com](mailto:bjoernsiepe@gmail.com)



<https://bsiepe.github.io>



[bsiepe.bsky.social](https://bsiepe.bsky.social)

## Slides & Papers



# References

- Efron, B. (2001). Statistical modeling: The two cultures: Comment. *Statistical Science*, 16(3), 218-219.
- Robinson, M.D., Vitek, O. (2019). Benchmarking comes of age. *Genome Biol* 20, 205 <https://doi.org/10.1186/s13059-019-1846-5>
- Rocca, R., & Yarkoni, T. (2021). Putting Psychology to the Test: Rethinking Model Evaluation Through Benchmarking and Prediction. *Advances in Methods and Practices in Psychological Science*, 4(3), 10.1177/25152459211026864. <https://doi.org/10.1177/25152459211026864>
- Sakaluk, J. K., & Short, S. D. (2016). A Methodological Review of Exploratory Factor Analysis in Sexuality Research: Used Practices, Best Practices, and Data Analysis Resources. *Journal of Sex Research*.
- Siepe, B. S., Haslbeck, J. M. B., Kloft, M., Büchner, A., Zhang, Y., Fried, E. I., & Heck, D. W. (2025). Introducing openESM: A database of openly available experience sampling datasets. [https://doi.org/10.31234/osf.io/qfdtb\\_v1](https://doi.org/10.31234/osf.io/qfdtb_v1)

# Resources

- All icons either from PowerPoint or fontawesome
  - <https://fontawesome.com/icons/bluesky>
  - <https://fontawesome.com/icons/glasses>
  - <https://fontawesome.com/icons/code>
  - <https://fontawesome.com/icons/dna>
- Python Software Foundation. (n.d.). *The Python logo*. <https://www.python.org/community/logos/> (PSF Trademark Policy)
- Rob Hyndman - <https://robjhyndman.com/hyndsight/m4comp/>
- Steni Sebastian - <https://medium.com/@ssteni/imagenet-what-why-and-how-e5ebed04abb5>
- R Foundation for Statistical Computing. (2016). *R logo*. <https://www.r-project.org/logo/> (CC-BY SA 4.0)
- James Grellier – Own Work, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=10253357>
- Marcin Wichary from San Francisco, U.S.A. - [1]Uploaded by Partyzan\_XXI, CC BY 2.0, <https://commons.wikimedia.org/w/index.php?curid=8198235>
- Carbon (<https://carbon.now.sh/>) for pretty code screenshots
- The Splintered Mind - <https://schwitzsplinters.blogspot.com/2021/02/adversarial-collaboration.html>
- Sandbox Studio Artwork - [https://www.symmetrymagazine.org/article/the-facts-and-nothing-but-the-facts?language\\_content\\_entity=und](https://www.symmetrymagazine.org/article/the-facts-and-nothing-but-the-facts?language_content_entity=und)
- Registered Reports graphic: Centre for Open Science, [https://cdn.cos.io/media/images/registered\\_reports.width-800.png](https://cdn.cos.io/media/images/registered_reports.width-800.png)

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